

Geometric Limits on Semantic Expressibility: Quantum-Inspired Contexts and Polysemy in Large Language Models

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Why can next-token prediction produce language that appears meaningful? Beginning with tokenization and vector encoding, we study two parts of this question. First, high-dimensional geometry permits exponentially many nearly orthogonal feature directions at a fixed tolerance, although their residual overlaps limit simultaneous readout. Second, polysemy requires us to distinguish static mixtures of word senses, context-dependent occurrence vectors, neural feature superposition, quantum coherence, and one direction shared by several orthonormal contexts. The quantum comparisons are structural and deliberately limited. We end with an informal explanation of self-attention, training, and autoregressive output; mathematical details are relegated to undergraduate-level appendices.

I. WHY DOES A SYNTACTIC MACHINE APPEAR TO UNDERSTAND?

A large language model (LLM) receives strings and returns strings. Yet it can answer questions, continue an argument, translate a passage, and alter its wording when the surrounding text changes. In this behavioral sense it often appears cognizant of what a text is about. How can a machine trained to predict linguistic form display abilities that we ordinarily describe in terms of meaning?

The question recalls John Searle’s Chinese Room thought experiment [1]. A person who does not know Chinese follows formal rules for transforming Chinese symbols and thereby produces answers that appear intelligent to an outside reader. Searle asks whether successful symbol manipulation is sufficient for understanding. We do not try to settle that philosophical question, and fluent behavior will not be used as evidence of consciousness. Our question is narrower and operational: what internal organization allows calculations on tokens to construct and exploit distinctions that human readers recognize as semantic?

There is an important clue. An LLM is not trained on arbitrary strings. Its corpus was produced by people writing about objects, actions, intentions, events, and one another. Regularities of use therefore contain indirect traces of regularities in human experience. The model is never handed a separate object called “meaning.” It is rewarded for predicting how language is used. Successful prediction can favor internal distinctions that track the contexts in which words and constructions occur. Whether this is intrinsic understanding or a powerful simulation remains open; the present chapter concerns the geometry of the representation.

Two issues provide a manageable route into that geometry.

First, a representation has a fixed number of independent coordinates, but it may need to distinguish vastly

more properties. High-dimensional vector spaces make this geometrically possible because they can contain very many directions that are not exactly orthogonal but have small mutual overlaps. We ask how this *quasi-orthogonality* enlarges expressive capacity and how accumulated overlap—or interference—limits access to the stored distinctions.

Second, one word can have several meanings. The word “bank” may refer to a financial institution, the edge of a river, or a seat. We ask how a single token type and its many occurrences can represent such polysemy. This leads to comparisons with two different quantum ideas: coherent superposition and one direction belonging to several orthonormal bases.

The order of exposition is intentional. Section II begins with tokenization and explains why vector representations are useful. Section III develops the abundance–interference tradeoff in informal terms. Section IV treats polysemy and the quantum comparisons. Only then does Sec. VI open the Transformer and explain attention, learning, and output generation. Formal statements and derivations appear in the appendices, where every symbol is defined as it is introduced.

II. FROM TEXT TO VECTORS

A. Tokenization: symbols before geometry

The first operation is *tokenization*. A tokenizer divides a character string into units drawn from a finite vocabulary and replaces each unit by an integer identifier. A token may be a complete word, part of a word, punctuation, a space-bearing fragment, or a byte-level unit. Thus a token is not automatically a word, still less a word’s meaning. Subword schemes assemble rare and previously unseen words from reusable fragments [2].

The assigned integer is merely an address. Identifier 7193 is not semantically closer to identifier 7194 than to identifier 20. The identifiers are therefore used as addresses in a table of vectors on which graded comparison

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and learned transformation are possible.

B. A token vector is a starting point, not a complete meaning

A learned lookup table assigns a real vector to every token identifier. Two occurrences of the same identifier begin with the same table row. The model must also know order: “the dog chased the cat” is not equivalent to “the cat chased the dog.” Position is therefore supplied in some form, either by adding positional information or by altering comparisons inside attention.

The resulting initial vector is not a contextually complete meaning. It has been trained across many occurrences and can reflect what those occurrences share, but later layers construct a new hidden vector for each particular occurrence. The occurrence of “bank” near “loan” can consequently acquire a different hidden state from the same token near “river,” although both began from the same lookup row. This distinction between a token *type* and a contextual token *occurrence* will be essential for polysemy.

C. Why vectors?

Vectors are useful for several reasons. Their coordinates can be adjusted continuously during learning. Matrices can transform them and inner products can compare them. Vector addition permits several contributions to coexist in one state. Most importantly here, lengths, directions, projections, and subspaces provide an economical language for graded features.

Suppose a unit vector u has been fitted as a direction associated with some property at a chosen layer. The scalar $u^T h$ is then the signed coordinate of a hidden state h along that direction. It is the coordinate, not the axis itself, that says on which side of the contrast the state lies and how strongly. In a simplified gender example, one may orient a fitted axis from a feminine-marked sample mean toward a masculine-marked sample mean. Positive and negative projections then describe opposite sides of the same fitted contrast; they are not two orthogonal directions. A midpoint or probe intercept must define the zero. Modern contextual representations may use several related directions rather than one universal axis, and the score can change from sentence to sentence [3, 4]. The superscript T means transpose, so $u^T h$ is the ordinary Euclidean inner product.

This directional interpretation is operational, not metaphysical. It depends on the layer, data, centering, inner product, and extraction method. A probe that reads a property does not by itself prove that the model uses the same direction causally. Nor does perpendicularity prove that two concepts are unrelated. Orthogonality has one precise consequence for a linear readout: a

component along one direction contributes no cross-talk when the other is read by an inner product.

The reason to emphasize vectors is therefore not that meanings are known in advance to be points in a Euclidean space. It is that a trained vector system can discover useful graded distinctions, combine them, and compare them by simple operations. High dimension adds a striking opportunity: far more approximately separated directions can coexist than there are exactly independent coordinates. That opportunity motivates our first central issue.

III. MANY DIRECTIONS IN A FIXED-WIDTH SPACE

A. Exact independence and approximate separation

In a d -dimensional space, no more than d nonzero vectors can be mutually orthogonal. This is elementary linear algebra: mutually orthogonal nonzero vectors are linearly independent, and a linearly independent family cannot contain more vectors than the dimension. If semantic distinctions required exactly perpendicular axes, fixed width would impose a severe limit.

Exact orthogonality is stronger than most readouts require. Let u and v be unit vectors. Their inner product is the cosine of the angle between them. We call them quasi-orthogonal at tolerance ε when $|u^T v| \leq \varepsilon$, where the tolerance is chosen before the comparison. They are not independent coordinates; they are approximately separated directions.

High-dimensional geometry makes such directions surprisingly abundant. For a uniformly random unit vector in d real dimensions, the overlap with a fixed unit direction is typically of size $1/\sqrt{d}$. Most of the sphere is therefore concentrated near the equator perpendicular to any chosen direction. Lévy’s concentration principle expresses this typicality. The Johnson–Lindenstrauss lemma approaches the same phenomenon from another side: a finite collection of points can be placed in a dimension proportional to the logarithm of the number of points while approximately preserving all pairwise distances. Read in the constructive direction, these results show that at any fixed nonzero angular tolerance a d -dimensional representation space admits at least exponentially many quasi-orthogonal directions [5–7].

The word *exponential* is important, but so are its qualifications. It counts directions available at the chosen tolerance, not a second vector-space dimension. It is a geometric possibility, not proof that a trained model uses all those directions. The allowed overlap must be stated, and an exponentially large dictionary is not automatically readable when many of its elements are active at once. Appendix B gives the concentration, union-bound, and Johnson–Lindenstrauss calculations.

B. The price of abundance: interference

The simplest calculation already shows the access problem. Suppose a hidden state contains two unit feature directions,

$$h = au + bv. \quad (1)$$

Reading the first feature by an inner product gives $u^\top h = a + b(u^\top v)$. The desired coefficient is a ; the second term is cross-talk from the other feature. Exact orthogonality removes it, while quasi-orthogonality bounds its size by $\varepsilon|b|$.

Many small unwanted terms can nevertheless accumulate. If k comparably sized, random-like features are active and their signs do not align systematically, their mean squared errors add as in the Pythagorean theorem. The resulting root-mean-square (RMS) interference is of order $\sqrt{k/d}$ relative to one unit-scale target. This is an average random-code benchmark, not a guarantee for every learned dictionary. In the worst case, aligned cross-terms can add rather than cancel.

The useful operating region is consequently task dependent. There should be enough distinguishable directions for the features a task needs, but not so many simultaneously active, strongly overlapping directions that the chosen readout fails. There is no universal numerical boundary determined by d alone. It also depends on which features coactivate, their strengths, the decoder, and the tolerated error. A learned model may use its angular budget selectively, placing greater separation between features that frequently need to be read together.

C. A limited analogy with geometric decoherence

Small overlaps also occur in quantum theory. When alternatives of a quantum system become correlated with nearly orthogonal environmental states, the off-diagonal terms of the reduced state are multiplied by the corresponding environmental overlaps. High dimension can make those factors very small when the conditional states or their relative dynamics satisfy an explicit typicality assumption; dimension alone is not enough, since the conditional states could remain identical. The author’s companion analysis studies this purely geometric contribution to decoherence [8].

The LLM analogy is narrow. Quasi-orthogonal semantic directions suppress linear cross-talk in much the way that nearly orthogonal environmental branch states suppress observable coherence terms. We call this *geometric decoherence*: unwanted contributions become small because the relevant inner products are small. It is not physical quantum decoherence. An LLM has no required system–environment split, unitary branch dynamics, partial trace, or local loss of access to phase relations after environmental degrees of freedom are traced out. High dimension supplies shared geometry, not a shared physical mechanism [9].

The word *orthogonal* also has different operational meanings here. For the LLM readout it means absence of linear cross-talk in the chosen inner product. Within one quantum measurement context, orthogonal rays represent mutually exclusive outcomes. The common inner-product geometry does not make these interpretations identical.

This qualified comparison is still useful. In both settings, exact orthogonality is an ideal limit; approximate orthogonality can already make alternatives operationally difficult to confuse; and the number of pairwise small overlaps is not the same as the number of alternatives that can be reliably recovered together.

IV. POLYSEMY: ONE FORM, SEVERAL MEANINGS

A. A sense is not a single feature

A *sense* is a contextually appropriate meaning of a lexical item. The financial, river-edge, and seat meanings of “bank” are different senses. The banking or tilting of an aircraft supplies another. A semantic feature is narrower: it may mark animacy, tense, number, sentiment, or one fitted gender contrast. Several features can contribute to one sense, and one feature can recur in many words and senses. Polysemy is therefore not solved merely by finding one direction for every dictionary definition.

It is also important to say which representation is being discussed. A static embedding assigns one stored vector to a word type. A Transformer has one initial lookup row for a token type but constructs different hidden vectors for its occurrences. These are related but not identical problems.

B. A static vector as a corpus-level mixture

In a static system such as word2vec, the same stored type vector is used in every occurrence. Under a particular generative model, Arora and collaborators establish an approximate two-sense relation between that vector and hypothetical sense vectors [10]. The following three-sense equation illustrates the corresponding finite-mixture idea:

$$\begin{aligned} v_{\text{bank}} &\approx p_{\text{fin}} v_{\text{fin}} + p_{\text{river}} v_{\text{river}} + p_{\text{seat}} v_{\text{seat}}, \\ p_{\text{fin}} + p_{\text{river}} + p_{\text{seat}} &= 1. \end{aligned} \quad (2)$$

The nonnegative coefficients summarize how frequently the senses occur in the corpus. The equation is an approximate classical convex combination and a population-level summary. It does not decide what one occurrence means, nor does one observed sum uniquely determine its hidden summands.

Arora et al. also expressed each word vector using only a few reusable, topic-like directions, which they called

discourse atoms. This sparse description can help recover recurring components across many words. It does not show that all sense vectors or atoms are mutually quasi-orthogonal, and the atoms need not correspond one-to-one with dictionary senses. The geometric results of Sec. III establish what is possible at a chosen tolerance; they do not establish what a learned lexical dictionary actually contains.

C. Contextual occurrences

Modern Transformers begin each occurrence of a token type from the same lookup row, apart from positional treatment, but the available sequence changes the subsequent calculation. The financial and river occurrences of “bank” can therefore acquire different hidden vectors. In a causal decoder, the state at a position uses only the prefix available there. If a disambiguating word follows “bank,” the earlier state for “bank” cannot see it, although later positions can combine both words. The phrase “surrounding context” must be understood with that causal qualification.

A simple interpretable model would let the available text reweight several sense vectors. A real Transformer is not required to implement that literal mixture. Its layers apply learned linear maps, nonlinear transformations, residual additions, and attention. Ambiguous sentences and puns may also preserve information relevant to more than one sense. It is therefore safer to say that contextualization *transforms* an occurrence representation than that it selects or collapses one pre-existing branch.

The phrase *neural feature superposition* denotes yet another classical idea. A hidden state may contain a sum of many feature directions, with a sparse subset appreciably active on a particular input [11, 12]. The coefficients may be signed and need not add to one; the features need not be word senses. Thus a static lexical mixture, a contextual occurrence state, and neural feature superposition are three different objects.

D. Why this is not automatically quantum coherence

Quantum theory uses the word *superposition* in a stronger operational sense. A pure state combines alternatives with probability amplitudes. Relative phase can affect later measurement probabilities when alternatives are recombined in another basis, and the Born rule turns amplitudes into probabilities. A classical frequency average such as Eq. (2) provides none of this. Its coefficients are frequencies, not amplitudes; no phase-sensitive transformation or interference experiment is specified; and its sense vectors are not declared orthonormal measurement outcomes.

Why not impose a full quantum-style superposition? One may certainly propose a new mathematical model

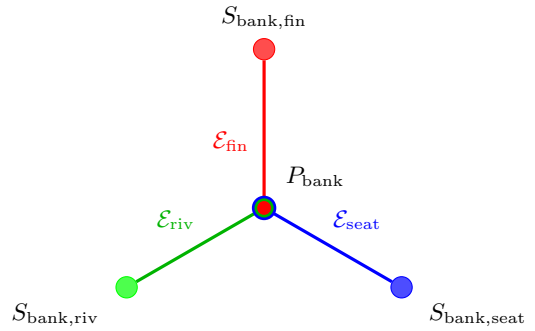


FIG. 1. Structural incidence schematic for “bank.” The central projector P_{bank} is shared. Each colored arm shows only a fragment of one context hyperedge; unshown orthogonal projectors complete the corresponding basis. An outer node $S_{\text{bank},s}$ summarizes a residual semantic sector, not necessarily one rank-one outcome. The drawing depicts shared incidence, not the vector sum in Eq. (2).

that does so. But the proposal would have to state the sense basis, amplitudes, relative phases, allowed transformations, measurement rule, and an experiment in which phase changes alter observable predictions. Without those ingredients, replacing the classical weights by symbols resembling quantum amplitudes merely renames a vector sum. Existing Transformer attention supplies no Born rule and no physical collapse. The explicit pure-state and density-matrix comparison is given in Appendix C.

E. A different quantum analogy: shared contexts

Quantum logic suggests a comparison that does not rely on vector addition. An orthonormal measurement context can be represented by mutually orthogonal rays, or equivalently by rank-one projectors that sum to the identity. In dimension at least three, distinct contexts may contain the same projector while differing in their other members. This is an incidence relation: one element belongs to several contexts.

The diagram below borrows that pattern for a speculative representation of “bank.” Its center is a fixed lexical projector. Each colored arm belongs to a different larger context, whose remaining directions carry context-specific information. The same lexical element is retained, while the financial, river-edge, and seat contexts differ around it.

This analogy has a sharp limit. The shared projector alone cannot say which sense is intended. If a quantum system were prepared exactly in that shared ray, a measurement in any basis containing it would return the shared outcome with certainty. More generally, for a fixed quantum state the Born probability of the same projector does not depend on which compatible projectors complete the basis. Linguistic interpretation is different in the relevant respect: it is supposed to depend on the

available text.

The diagram can therefore motivate only an optional classical hypothesis: a contextual occurrence vector might retain a stable nonzero lexical component while placing sense-dependent information in accompanying directions. Such a decomposition is mathematically easy to construct; its scientific content would lie in stability, predictive usefulness, and improvement over simpler untied contextual models. The analogy is structural, not a claim that polysemy is a quantum measurement. Appendix C gives the shared-projector construction and explains why quantum probability rules do not transfer to linguistic context.

V. WHAT IS ESTABLISHED, WHAT IS PROPOSED, AND HOW IT COULD FAIL

This chapter reports no new experiment. It combines established mathematics with empirical hypotheses, and that distinction should remain visible.

The geometric statements are established. At a fixed nonzero tolerance, high-dimensional spheres admit exponentially large families of directions with bounded pairwise overlap. Uniformly random pairs have RMS overlap $1/\sqrt{d}$. For a specified linear correlation readout, overlap produces the cross-term displayed in Eq. (1). These facts say nothing by themselves about which semantic features an LLM has learned.

The first empirical question is therefore whether feature directions extracted by a declared method show useful geometry on held-out data. Their overlaps should be compared with isotropic random controls and with activity-matched controls. The most relevant overlaps are those between features that coactivate and need to be read together. Failure to beat these baselines would weaken the claim that learning has allocated angular separation strategically.

The second question concerns polysemy. An Arora-style static mixture should be tested in one declared lexical space with sense frequencies estimated from training data. A contextual representation should then predict held-out senses or same-sense judgments better than that static prior. Otherwise the contextual account has added complexity without demonstrated benefit.

The third question concerns the optional shared anchor in Fig. 1. It must be learned on training data, remain non-negligible and stable on held-out occurrences, and improve prediction, compression, or sample efficiency relative to equally flexible models without the shared-anchor constraint. Since any chosen vector can be completed to an orthonormal basis, merely drawing several contexts around one ray is no evidence. If the anchor becomes negligible, varies strongly with context, or fails against simpler controls, this optional proposal should be rejected while the ordinary contextual account remains intact.

No such test establishes consciousness, intrinsic inten-

tionality, or a physical quantum process. The measurable claims concern representation, overlap, readout, and predictive behavior.

VI. HOW ATTENTION USES CONTEXT AND PRODUCES TEXT

We have so far treated the Transformer—a layered neural-network architecture centered on attention—as a device that changes an initial token vector into a context-dependent occurrence vector. We now return to that missing mechanism. The informal account needs only vectors, matrices, inner products, weighted averages, and smooth nonlinear transformations.

A. Two meanings of “weight”

The word *weight* is used for two different things. The embedding table and the matrices inside attention, feed-forward, and output blocks are *persistent model parameters*. Training changes them, and the resulting numbers survive from one prompt to the next. Self-attention also computes temporary, input-dependent numbers commonly called attention weights. To avoid confusion, we call the latter *attention coefficients*. They are recomputed for every sequence, layer, head, and permitted target–source pair of positions. Attention does not itself create the persistent model parameters.

B. Self-attention as context-dependent routing

At a given layer, every token position carries a current hidden vector. One attention head applies three learned linear maps to these vectors. Their outputs are conventionally called queries, keys, and values. The query and key are two transformed views used to calculate a match score; the value is the transformed content routed according to that score. The names are mnemonics rather than literal mental operations.

For one target position, the head compares its query with the keys at the positions it is allowed to inspect. Inner products supply the comparison scores. A text-generating, decoder-only model uses a causal mask, so it cannot inspect future tokens. Softmax normalization turns the permitted scores into nonnegative attention coefficients whose sum is one. The head then forms a weighted average of the corresponding value vectors [13].

This weighted average is only the immediate output of one head. A practical layer combines several heads. An output map combines their results; a residual connection adds the layer’s input back; and a feed-forward block is a small nonlinear network applied at each position. The complete layer is therefore neither a mere average nor an orthogonal projection. Repeated layers allow the

representation at one position to gather and transform information from many permitted positions.

For “bank,” evidence such as “deposit,” “loan,” and “interest” can create different query–key matches and value contributions than “river,” “water,” and “shore.” At the position of “bank,” a causal model can use only evidence already in the prefix; evidence appearing later can be combined with “bank” only at later positions. Attention thereby participates in contextualization and may retain, combine, or later discard several kinds of evidence.

C. How training learns the persistent parameters

During language-model training, the network sees a token sequence and is asked at each permitted position to predict the following token. One sequence therefore supplies many prediction examples: the state at position i predicts the token at position $i + 1$. Because the causal mask prevents access to later tokens, the eligible distributions can be computed in parallel during one forward pass. A loss is large when the continuation that actually occurred received little probability and small when it received much probability.

Backpropagation applies the multivariable chain rule to determine how the loss changes when each persistent parameter changes. An optimizer then adjusts the parameters slightly to lower the current minibatch loss. Repeating the forward–loss–backward–update cycle over many varied text segments aims to improve prediction on unseen continuations and produces the trained model [14].

Self-attention matters during this process because the error is differentiated through its query, key, value, and softmax calculations. Nevertheless, in each individual forward pass the persistent matrices are fixed. Attention produces contextual activations and temporary coefficients; gradient-based optimization produces the trained parameter values.

D. Autoregressive output

When a trained model answers a prompt, its persistent parameters normally remain fixed. The prompt is tokenized, mapped to initial vectors, and passed through the Transformer. An output map converts the final-layer hidden state at the last position of the current prefix into one score for every token in the vocabulary. Softmax turns those scores into a distribution over possible next tokens; it is not a probability that the completed statement is true. A decoding rule either chooses a high-scoring token or samples from a selected portion of that distribution [15].

The chosen identifier is appended to the prefix. The enlarged prefix becomes the input for the following prediction, so one generated token can change all later hidden

states and attention coefficients. The model ordinarily does not compose a complete answer in a hidden store and then reveal it. It constructs the response one token at a time. Finally, detokenization joins the generated pieces into readable text.

This loop is recursive in its state but not usually in its persistent parameters: output becomes new input, while the deployed weights remain unchanged. External memory or retrieval can also change later inputs without changing the model. Genuine online learning would be a separate parameter-update procedure, not ordinary autoregression.

The calculation now closes the conceptual loop. The model never receives a dictionary entry called “meaning.” It receives token occurrences in human text, turns them into vectors, and learns transformations that improve prediction. Some of the resulting geometry supports distinctions and contextual responses that readers interpret semantically. This explains a plausible, testable route to meaning-sensitive behavior without deciding whether the machine understands in Searle’s philosophical sense.

VII. CONCLUSION

Searle’s Chinese Room asks whether correct manipulation of symbols amounts to understanding. We have not answered that philosophical question. We have instead followed one operational path from tokens to meaning-sensitive behavior.

Token identifiers first become vectors because vectors support graded coordinates, learned transformations, comparison, and combination. Their high-dimensional geometry addresses the first problem considered here. A fixed-width representation space can accommodate exponentially many quasi-orthogonal directions at a chosen tolerance, even though it has only d exactly independent coordinates. This is expressive room, not cost-free access. Residual overlaps create interference, especially when many relevant features are active together.

The resulting overlap suppression has a limited quantum counterpart. Small inner products suppress cross-terms both in linear semantic readout and in the environmental overlaps that enter quantum decoherence. The shared geometry justifies the phrase geometric decoherence, while the absence of quantum dynamics, relative phases, and environmental tracing prevents a physical identification.

Polysemy exposes different limits of the quantum comparison. A static word vector may approximate a frequency-weighted convex mixture of sense vectors, and a Transformer constructs context-dependent occurrence vectors. Neither is thereby a coherent quantum superposition, and attention is not collapse. The star diagram retains only a quantum-logical incidence idea: one lexical projector may be shared by several orthonormal contexts. That optional constraint must earn its place empirically; the common ray alone does not select a meaning.

Attention finally explains how available tokens affect one another’s representations, training explains how prediction errors change persistent parameters, and autoregression explains how vectors are converted back into a sequence of tokens. Together these mechanisms outline a plausible route by which syntactic prediction can acquire a geometry useful for semantic discrimination. Whether useful semantic discrimination is understanding remains the question with which we began.

Appendix A: Formal companion to *From text to vectors*

This appendix gives a minimal mathematical version of Sec. II. It distinguishes strings, token identifiers, and vectors before discussing semantic directions.

1. Tokenization and lookup

Let $\mathcal{S}$ be the set of finite character strings. Let $\mathcal{V} = \{1, \dots, V\}$ be a vocabulary containing V token identifiers. Each identifier is an integer label. A tokenizer is a fixed map

$$\text{Tok} : \mathcal{S} \longrightarrow \bigcup_{T \geq 0} \mathcal{V}^T. \quad (\text{A1})$$

If $\text{Tok}(\sigma) = (t_1, \dots, t_T)$, then the input string σ has become a sequence of T identifiers. The union in Eq. (A1) allows strings to produce sequences of different lengths. A detokenizer reverses the assembly, subject to the system’s stated normalization rules.

Let d_{model} be the architectural width of the model’s hidden stream. The notation $\mathbb{R}^m$ means the set of real column vectors with m coordinates. An embedding table is a real matrix $E \in \mathbb{R}^{V \times d_{\text{model}}}$. Its row $E[t_i]$ is the initial lookup vector for identifier t_i . In a simple additive positional scheme,

$$h_i^{(0)} = E[t_i] + p_i, \quad h_i^{(0)}, p_i \in \mathbb{R}^{d_{\text{model}}}. \quad (\text{A2})$$

Here p_i supplies the position and the superscript (0) means the stage before the first Transformer layer. Some architectures encode position by rotating queries and keys rather than literally adding p_i ; the conceptual role is the same. The identifier t_i , the table row $E[t_i]$, and the contextual state produced at a later layer are three different objects.

The geometric analysis need not use all d_{model} architectural coordinates. A declared preprocessing map may center the observed states and remove nearly unused directions. It may also rescale the retained principal directions to a declared common variance, a step called whitening. We use $d \leq d_{\text{model}}$ for the rank of the resulting representation space. Any numerical claim involving d must therefore report the model, layer, sample, centering, rank threshold, and inner product used to obtain it.

2. Projection onto a fitted feature direction

Let $u \in \mathbb{R}^d$ be a unit vector, so $u^\top u = 1$. The orthogonal projection of a state h onto the line through u is

$$P_u h = (u^\top h)u, \quad P_u = uu^\top. \quad (\text{A3})$$

The scalar $u^\top h$ is its signed coordinate. The matrix P_u is called a projector because direct multiplication gives $P_u^2 = P_u$. The remainder $h - P_u h$ is perpendicular to u , since $u^\top (h - P_u h) = 0$. This is the elementary orthogonal projection theorem in finite-dimensional form.

An oriented contrast needs a baseline. Suppose two training samples have means μ_A and μ_B in the same representation space and $\mu_A \neq \mu_B$. Define

$$g = \frac{\mu_A - \mu_B}{\|\mu_A - \mu_B\|}, \quad h_0 = \frac{\mu_A + \mu_B}{2}, \quad (\text{A4})$$

$$\alpha_g(h) = g^\top (h - h_0).$$

The unit vector g orients the contrast, h_0 declares its midpoint, and $\alpha_g(h)$ is the signed coordinate. Substitution shows

$$\alpha_g(\mu_A) = \frac{\|\mu_A - \mu_B\|}{2}, \quad \alpha_g(\mu_B) = -\frac{\|\mu_A - \mu_B\|}{2}. \quad (\text{A5})$$

Swapping the two groups reverses g and every sign but does not change the unoriented line. A score near zero means that the coordinate along g is near the midpoint; it does not imply that the full vector is near h_0 , because a large perpendicular component may remain.

The same construction can illustrate a fitted gender-associated contrast, but it does not define gender and need not capture all gender-related information. A linear classifier provides a related alternative: if its decision boundary is $w^\top h + \beta_0 = 0$ with $w \neq 0$, then $(w^\top h + \beta_0)/\|w\|$ is signed Euclidean distance from that hyperplane. Either construction must be fitted on training examples and evaluated on held-out examples.

Appendix B: Formal companion to the geometric argument

The main text used three counts. The integer d is the exact dimension of the selected vector space. The integer M is the number of feature or code directions in a declared dictionary. The integer k is the number of those directions appreciably active in one state. Only d is a vector-space dimension.

1. Exact orthogonality and quasi-orthogonality

Let $w_1, \dots, w_M \in \mathbb{R}^d$ be unit vectors. The family is ε -quasi-orthogonal when

$$|w_i^\top w_j| \leq \varepsilon \quad \text{whenever } i \neq j, \quad 0 < \varepsilon < 1. \quad (\text{B1})$$

The absolute value treats a direction and its negative as the same unoriented axis for purposes of overlap.

When $\varepsilon = 0$, the vectors are mutually orthogonal. To see why there can then be at most d , suppose $a_1 w_1 + \dots + a_M w_M = 0$. Taking the inner product with w_j leaves $a_j = 0$, because every cross-term vanishes and $w_j^\top w_j = 1$. All coefficients are zero, so the vectors are linearly independent. The definition of dimension now gives $M \leq d$.

For $\varepsilon > 0$, the vectors need not be linearly independent. The largest possible M becomes a finite spherical-code problem at the chosen tolerance. Its cardinality is sometimes informally called quasi-dimensionality, but no additional linear dimensions have appeared.

2. Why typical overlaps shrink with dimension

For a random quantity Y , $\mathbb{E}[Y]$ denotes its average. For an event A , $\mathbb{P}(A)$ denotes its probability. A unit vector chosen uniformly from the sphere has no preferred direction: rotating the whole experiment does not change its probability law.

Fix a unit vector u and choose X uniformly from the unit sphere in $\mathbb{R}^d$. Rotational symmetry lets us take u to be the first coordinate direction. The overlap $u^\top X$ is then X_1 . Symmetry makes all coordinate squares have the same expectation. Since $X_1^2 + \dots + X_d^2 = 1$, taking expectations gives

$$1 = d\mathbb{E}[X_1^2], \quad \mathbb{E}[(u^\top X)^2] = \frac{1}{d}. \quad (\text{B2})$$

The RMS overlap is the square root of the mean squared overlap, hence $1/\sqrt{d}$. This elementary calculation gives a scale, not a tail probability and not a guarantee for learned vectors.

Lévy’s concentration lemma supplies the tail. One convenient consequence is that universal positive constants c and C exist such that

$$\mathbb{P}\{|u^\top X| > \varepsilon\} \leq C \exp(-cd\varepsilon^2) \quad (\text{B3})$$

for a fixed unit u and a relevant fixed tolerance. The function $x \mapsto u^\top x$ is 1-Lipschitz because the Cauchy–Schwarz inequality gives $|u^\top(x - y)| \leq \|x - y\|$. Lévy’s lemma says that a Lipschitz function on a high-dimensional sphere is unlikely to deviate far from its typical value. Here 1-Lipschitz means that changing the input by a distance r changes the function value by at most r [5, 16].

3. From one pair to an exponential family

Choose M directions independently and uniformly. There are $M(M-1)/2$ unordered pairs. The union bound states that the probability of at least one bad pair is no

larger than the sum of the bad-pair probabilities. Combining it with Eq. (B3) gives

$$\mathbb{P}\left\{\max_{i \neq j} |w_i^\top w_j| > \varepsilon\right\} \leq \frac{CM(M-1)}{2} \exp(-cd\varepsilon^2). \quad (\text{B4})$$

Choose the greatest integer M not exceeding $\exp(c'\varepsilon^2 d)$, where c' is sufficiently small and d is sufficiently large. Then the right side is below one. Therefore at least one family of that exponential size has every overlap within tolerance. This is an existence lower bound and a random-code benchmark, not a universal maximum and not proof that an LLM learns such a code.

Solving the exponent in Eq. (B4) for the overlap tolerance shows that the union-bound crossover is a constant multiple of $\sqrt{\log M/d}$ when $\log M$ is small compared with d . The union bound supplies a high-probability upper scale, not by itself a matching lower law. This remains a null-model estimate rather than a hard capacity wall.

4. The Johnson–Lindenstrauss route

The Johnson–Lindenstrauss lemma states that a set of $M+1$ points can be mapped into $\mathbb{R}^d$ while preserving all squared distances within relative error δ when

$$d \geq C_{\text{JL}} \delta^{-2} \log(M+1), \quad 0 < \delta < 1, \quad (\text{B5})$$

where C_{JL} is a numerical constant and $\log$ is the natural logarithm [6, 7]. This is a theorem about a prescribed finite set, not random semantic features.

To obtain a quasi-orthogonal code, take the origin and M mutually orthogonal standard coordinate vectors e_i in a sufficiently large starting space. Let Φ be a linear Johnson–Lindenstrauss map and define $y_i = \Phi e_i$. Including the origin controls the image lengths. The polarization identity

$$2y_i^\top y_j = \|y_i\|^2 + \|y_j\|^2 - \|y_i - y_j\|^2 \quad (\text{B6})$$

turns preservation of lengths and distances into preservation of inner products. Before normalization their pairwise absolute inner products are at most 2δ ; because every squared length is at least $1 - \delta$, the normalized overlaps are at most $2\delta/(1 - \delta)$. For a fixed target dimension d , choose an integer M satisfying $M+1 \leq \exp(c\delta^2 d)$, with c small enough that Eq. (B5) holds. This constructs an exponential-size family.

Lévy concentration, random spherical coding, and the Johnson–Lindenstrauss lemma are thus related views of one high-dimensional phenomenon. Their exponential factors must not be multiplied as though they were independent mechanisms.

5. A deterministic overlap floor

Random coding gives a typical construction. The Welch bound gives a deterministic limitation. Put the

vectors into the columns of a matrix $W \in \mathbb{R}^{d \times M}$ and form the Gram matrix $G = W^\top W$. Its entry $G_{ij} = w_i^\top w_j$ records every pairwise overlap. The diagonal entries are one, so the trace of G is M . The matrix has rank at most d and nonnegative eigenvalues.

Let $\lambda_1, \dots, \lambda_r$ be its nonzero eigenvalues, where $r \leq d$. Cauchy–Schwarz gives

$$M^2 = \left(\sum_{\ell=1}^r \lambda_\ell \right)^2 \leq r \sum_{\ell=1}^r \lambda_\ell^2 \leq d \sum_{\ell=1}^r \lambda_\ell^2. \quad (\text{B7})$$

The sum of squared matrix entries equals the sum of squared eigenvalues for a real symmetric matrix. Removing the M diagonal ones and averaging the remaining entries yields

$$\max_{i \neq j} |w_i^\top w_j|^2 \geq \frac{M-d}{d(M-1)} \quad (M > d). \quad (\text{B8})$$

This is the Welch bound [17]. It says that once the dictionary is overcomplete, some nonzero overlap is unavoidable. It does not say which pair overlaps or how damaging that pair is for a particular task.

6. Two-feature and many-feature readout

Let a hidden state be represented by

$$h = \sum_{i \in A} a_i w_i + e. \quad (\text{B9})$$

Here A is the active set, a_i are scalar feature coefficients, and e contains components omitted by this chosen dictionary. Reading feature $j \in A$ by correlation gives

$$w_j^\top h = a_j + \sum_{\substack{i \in A \\ i \neq j}} a_i (w_j^\top w_i) + w_j^\top e. \quad (\text{B10})$$

The first term is desired; the sum contains geometric cross-talk. Equation (1) is the case $A = \{1, 2\}$ and $e = 0$.

If the directions are ε -quasi-orthogonal, the triangle inequality gives the worst-case bound

$$\left| \sum_{\substack{i \in A \\ i \neq j}} a_i (w_j^\top w_i) \right| \leq \varepsilon \sum_{\substack{i \in A \\ i \neq j}} |a_i|. \quad (\text{B11})$$

This bound allows every unwanted term to have the same sign. Under a different, explicitly random-like model, assume instead that the cross-terms are centered, have zero cross-moments, and satisfy $\mathbb{E}[(w_j^\top w_i)^2 | w_j] = 1/d$. Treat the active set and coefficients as fixed and condition on w_j . Variances then add:

$$\mathbb{E} \left[\left(\sum_{\substack{i \in A \\ i \neq j}} a_i w_j^\top w_i \right)^2 \middle| A, (a_i), w_j \right] = \frac{1}{d} \sum_{\substack{i \in A \\ i \neq j}} a_i^2. \quad (\text{B12})$$

For $k-1$ interfering features of comparable unit coefficient, the square root is $\sqrt{(k-1)/d}$. This is the RMS law quoted in the main text. Centering or zero cross-moments is essential; an informal claim of uncorrelated contributions would be insufficient without their means being defined.

The contrast between Eqs. (B11) and (B12) explains why no universal Goldilocks boundary follows from d and M alone. A task must also specify active coefficients, coactivation frequencies, a decoder, and an accepted error. Overlap between two features that never need to be recovered together may have little cost.

7. The common algebra behind geometric decoherence

We first introduce the notation used in this comparison. A quantum state is a complex column vector. Dirac notation writes such a vector as a *ket* $|v\rangle$ and its conjugate transpose as the *bra* $\langle v|$. Thus $\langle v|w\rangle$ is an inner product, whereas $|v\rangle\langle w|$ is a matrix called an outer product. A density matrix is a positive semidefinite matrix of trace one; the trace is the sum of its diagonal entries. The partial trace over an environment is the linear operation that discards the environmental coordinates while preserving all expectation values of measurements made only on the system.

For comparison with physical decoherence, let $\{|s_i\rangle\}$ be orthonormal alternatives of a quantum system. Suppose a system–environment interaction produces the pure joint state

$$|\Psi\rangle = \sum_i c_i |s_i\rangle |E_i\rangle, \quad \sum_i |c_i|^2 = 1. \quad (\text{B13})$$

The normalized $|E_i\rangle$ are conditional environmental states. Observing only the system means taking a partial trace over the environment. The reduced system state is

$$\rho_S = \sum_{i,j} c_i c_j^* \langle E_j | E_i \rangle |s_i\rangle \langle s_j|. \quad (\text{B14})$$

Thus every off-diagonal term with $i \neq j$ is multiplied by an environmental overlap. When the relevant $|E_i\rangle$ become nearly orthogonal, those coherence terms become small for system-only observations. This is the standard algebraic core of environmental decoherence [9].

Near-orthogonality of the $|E_i\rangle$ requires a stated assumption about the interaction or branch-state ensemble; high dimension alone does not produce it. Moreover, many small off-diagonal terms can accumulate. Therefore, bounding each pair separately does not by itself show that the whole reduced matrix is close to the matrix obtained by deleting all off-diagonal terms.

Compare Eq. (B14) with Eq. (B10). The quantum coherence term is suppressed by $\langle E_j | E_i \rangle$; the classical readout error is suppressed by $w_j^\top w_i$. Both are small-overlap effects. Only the former follows from a joint quantum

state and a partial trace. The latter is ordinary cross-talk in a real linear readout. This is why geometric decoherence is an analogy in the LLM setting rather than a claim of physical decoherence.

8. Conditional tensor-product amplification

One further scaling is mathematically correct but not implied for an ordinary Transformer. If a separate model genuinely has n tensor factors, each of fixed local dimension $q \geq 2$, then its exact Hilbert-space dimension is

$$d = q^n. \quad (\text{B15})$$

Substitution into the exponential code bound gives

$$M \geq \exp(c\varepsilon^2 q^n) - 1, \quad (\text{B16})$$

which is doubly exponential in the number of factors n at fixed q and ε , for sufficiently large n . This is a code cardinality, not an additional Hilbert-space dimension. A real code is also a valid subset of the corresponding complex representation space. Standard LLM residual width is not known to equal an accessible q^n tensor-product representation space, so Eq. (B16) is not an ordinary LLM capacity claim.

Appendix C: Formal companion to the polysemy comparison

1. Static mixture, contextual reweighting, and feature code

Let t denote a word type, or a token that corresponds to a complete lexical item, and let $s \in \{1, \dots, S_t\}$ index its candidate senses. Let $v_{t,n}^{\text{type}}$ be the static embedding estimated from an amount of corpus data indexed by n . Under the generative assumptions used by Arora et al., this estimate converges in the corpus limit to a population vector $\bar{v}_t$. That limit is approximately related to sense-specific embeddings $q_{t,s}$ by

$$\begin{aligned} v_{t,n}^{\text{type}} &\longrightarrow \bar{v}_t & (n \rightarrow \infty), \\ \bar{v}_t &\approx \sum_{s=1}^{S_t} \pi_t(s) q_{t,s}, & \sum_s \pi_t(s) = 1. \end{aligned} \quad (\text{C1})$$

If $f_{t,s}$ is the corpus count of sense s among occurrences of type t , then $\pi_t(s) = f_{t,s} / \sum_r f_{t,r}$ is its relative frequency. The first relation is the asymptotic convergence statement; the second is the model-dependent approximation to a frequency-weighted sense sum. Arora et al. prove the displayed relation explicitly for two senses; the finite multisense expression is their general linearity assertion and follows from the same conditioning argument [10]. Keeping the two relations separate avoids attributing every modeling residual to finite-sample error.

An interpretable contextual extension replaces the fixed frequency prior by weights inferred from the available text x :

$$\begin{aligned} q_t^{\text{mix}}(x) &= \sum_{s=1}^{S_t} p_\theta(s | x, t) q_{t,s}, \\ p_\theta(s | x, t) &\geq 0, & \sum_s p_\theta(s | x, t) &= 1. \end{aligned} \quad (\text{C2})$$

The subscript θ denotes fitted parameters of the contextual weighting rule. Equation (C2) is a useful classical baseline. An actual Transformer guarantees only that some learned map F_ℓ produces an occurrence state

$$h_\ell(t | x) = F_\ell(E[t], x) \quad (\text{C3})$$

at layer ℓ . It need not implement the explicit convex mixture in Eq. (C2).

A general neural feature code has still another form:

$$h_\ell(t | x) = b_\ell + \sum_{i=1}^M a_i(t, x) w_i + e_\ell(t, x). \quad (\text{C4})$$

Here b_ℓ is a baseline, w_i are learned or extracted feature directions, a_i are input-dependent activations, and e_ℓ is the unmodeled residual. The a_i may be signed and need not sum to one. Only a subset may be appreciably active. Thus Eq. (C4) is a linear decomposition of an activation, not a probability mixture of word senses.

2. Quantum pure superposition and incoherent mixture

Using the Dirac notation introduced in Appendix B 7, let $\{|s\rangle\}$ now denote orthonormal quantum alternatives. A normalized pure state is

$$|\psi\rangle = \sum_s c_s |s\rangle, \quad \sum_s |c_s|^2 = 1. \quad (\text{C5})$$

The coefficients c_s are amplitudes. They may be real or complex; quantum coherence does not require every amplitude to be non-real. What matters is that relative signs or phases can affect probabilities after a change of measurement basis.

The density matrix of the pure state is

$$|\psi\rangle\langle\psi| = \sum_{s,r} c_s c_r^* |s\rangle\langle r|. \quad (\text{C6})$$

The star denotes complex conjugation. Terms with $s \neq r$ are off-diagonal coherences in this basis. By contrast, an incoherent quantum state formed by classical probabilistic mixing of the same orthogonal alternatives is represented by

$$\rho_{\text{mix}} = \sum_s p_s |s\rangle\langle s|, \quad p_s \geq 0, \quad \sum_s p_s = 1. \quad (\text{C7})$$

This positive semidefinite, trace-one matrix has no off-diagonal terms in the displayed basis. Under its intended lexical interpretation, the Arora sum in Eq. (C1) is neither Eq. (C5) nor Eq. (C7): it is a real vector average in a lexical embedding space. Ordinary Euclidean cross-terms do not establish quantum interference.

3. The optional shared-projector construction

Let $r_t \in \mathbb{R}^d$ be a unit lexical anchor and let

$$P_t = r_t r_t^\top \quad (\text{C8})$$

be its rank-one projector. For linguistic context class s , choose m_s orthonormal residual directions perpendicular to r_t and place them in the columns of

$$\begin{aligned} B_{t,s} &= [u_{t,s,1} \cdots u_{t,s,m_s}], \\ B_{t,s}^\top B_{t,s} &= I_{m_s}, \quad B_{t,s}^\top r_t = 0. \end{aligned} \quad (\text{C9})$$

The identity matrix I_{m_s} has m_s rows and columns. The columns of $B_{t,s}$ form an orthonormal partial frame. The Gram–Schmidt or orthonormal-extension theorem permits it to be completed to a full basis of $\mathbb{R}^d$. Put $P_{t,s,j+1} = u_{t,s,j} u_{t,s,j}^\top$ for the displayed residual directions and use the additional basis directions for the remaining projectors. In projector language, a completed context has the form

$$\mathcal{E}_{t,s} = \{P_t, P_{t,s,2}, \dots, P_{t,s,d}\}, \quad P_t + \sum_{j=2}^d P_{t,s,j} = I_d. \quad (\text{C10})$$

The $P_{t,s,j}$ are mutually orthogonal rank-one projectors. Different hyperedges $\mathcal{E}_{t,s}$ may share P_t and differ in their other projectors. The orthonormal completion is nonunique, so its mere existence is automatic mathematics rather than an empirical discovery.

A strict shared-anchor hypothesis for contextual states could be written

$$\begin{aligned} h_\ell(t \mid x) &= a_t r_t + B_{t,s(x)} z_{t,s(x)}(x) \\ &\quad + e_t(x), \\ P_t e_t(x) &= 0, \\ B_{t,s(x)}^\top e_t(x) &= 0. \end{aligned} \quad (\text{C11})$$

Here $s(x)$ is a context class inferred from text, $z_{t,s(x)}(x)$ contains context-dependent residual coordinates, and $a_t \neq 0$ is required to be the same across contexts. The two displayed orthogonality conditions put $e_t(x)$ outside both the anchor and the selected residual sector, so components cannot be shifted arbitrarily between $B_{t,s(x)} z_{t,s(x)}(x)$ and $e_t(x)$. Only the component $a_t r_t$ is fixed; the complete occurrence vector generally changes. If the entire vector were fixed, the context would have to be stored elsewhere and that vector alone would not encode polysemy.

For a fixed quantum state ρ , the Born probability assigned to the shared projector is

$$p(P_t \mid \rho) = \text{tr}(\rho P_t). \quad (\text{C12})$$

No term in this expression names the other projectors that complete the measurement context. Gleason’s theorem, under its additivity hypotheses in real or complex dimension at least three, derives this density-operator form for projector probabilities [18]. This limited probability consistency does not say that quantum mechanics admits a global noncontextual assignment of sharp values. The Kochen–Specker theorem rules out such assignments under its own assumptions [19, 20]. Neither theorem implies that linguistic context should obey a Born rule.

Appendix D: Formal companion to attention, training, and output

Let Θ denote all persistent model parameters: the embedding table, attention and feed-forward matrices, normalization parameters, and vocabulary readout. During one forward pass Θ is fixed. Hidden states, queries, keys, values, and attention coefficients are transient quantities computed from the current sequence.

1. One causal attention head

Let $h_i \in \mathbb{R}^{d_{\text{model}}}$ be the current hidden vector at target position i . Let d_h be the width of one attention head. Three learned matrices $W_Q, W_K, W_V \in \mathbb{R}^{d_h \times d_{\text{model}}}$ produce

$$q_i = W_Q h_i, \quad k_j = W_K h_j, \quad v_j = W_V h_j. \quad (\text{D1})$$

The vectors q_i, k_j , and v_j are the query at target position i and the key and value at source position j .

In a decoder-only causal model, position i may use positions $j \leq i$. The scaled score, attention coefficient, and one-head aggregate are

$$\begin{aligned} s_{ij} &= \frac{q_i^\top k_j}{\sqrt{d_h}}, \\ \alpha_{ij} &= \frac{\exp(s_{ij})}{\sum_{r=1}^i \exp(s_{ir})}, \\ a_i^{\text{att}} &= \sum_{j=1}^i \alpha_{ij} v_j. \end{aligned} \quad (\text{D2})$$

The factor $1/\sqrt{d_h}$ keeps the scale of the scores roughly stable. If coordinate products have comparable variance, adding d_h of them makes an unscaled dot product’s typical size grow like $\sqrt{d_h}$. The exponential is positive, so every $\alpha_{ij} \geq 0$, and direct summation shows $\sum_{j=1}^i \alpha_{ij} = 1$. Thus the immediate aggregate of one head is a convex

combination of its permitted value vectors. The complete multi-head block subsequently applies output maps and a residual connection, so the whole Transformer layer need not be a convex combination or a projector.

The Cauchy–Schwarz inequality gives $|q_i^\top k_j| \leq \|q_i\| \|k_j\|$. A large score therefore requires appropriate alignment and scale. The learned matrices decide which aspects of the incoming states become relevant to the comparison.

2. Next-token probabilities and training

After the final layer L , let $h_i^{(L)}$ be the contextual state at position i . Let $W_{\text{out}} \in \mathbb{R}^{V \times d_{\text{model}}}$ and $b \in \mathbb{R}^V$ be the vocabulary readout matrix and bias. The score vector $g_i \in \mathbb{R}^V$ and the probability of candidate token $a \in \mathcal{V}$ are

$$g_i = W_{\text{out}} h_i^{(L)} + b, \quad p_{\Theta}(a \mid t_1, \dots, t_i) = \frac{\exp(g_{i,a})}{\sum_{c \in \mathcal{V}} \exp(g_{i,c})}. \quad (\text{D3})$$

The entries of g_i are scores, often called logits. The softmax denominator makes the probabilities nonnegative and makes them sum to one over the vocabulary.

For a training sequence $(t_1, \dots, t_T)$ with $T \geq 2$, the standard next-token negative log-likelihood is

$$\mathcal{L}(\Theta) = - \sum_{i=1}^{T-1} \log p_{\Theta}(t_{i+1} \mid t_1, \dots, t_i). \quad (\text{D4})$$

Each term is small when the observed next token received high probability. Backpropagation applies the multivariable chain rule to compute the gradient $\nabla_{\Theta} \mathcal{L}$. The elementary gradient step

$$\Theta_{\text{new}} = \Theta - \eta \nabla_{\Theta} \mathcal{L}, \quad \eta > 0, \quad (\text{D5})$$

changes the persistent parameters. At a point where the loss is differentiable, the first-order Taylor formula explains the minus sign:

$$\mathcal{L}(\Theta - \eta \nabla \mathcal{L}) \approx \mathcal{L}(\Theta) - \eta \|\nabla \mathcal{L}\|^2. \quad (\text{D6})$$

For a nonzero gradient and a sufficiently small step, the linear term lowers the same loss locally. Practical optimizers add batching, adaptive scaling, momentum, regularization, and other refinements; the separation remains:

attention computes transient coefficients, while optimization updates persistent parameters.

3. Autoregressive generation

Let $(u_1, \dots, u_m)$ be the tokenized prompt, where m is its length, and let Θ^* be the deployed parameters. Here the star labels the chosen deployment checkpoint; it is not complex conjugation. After processing a current prefix $(u_1, \dots, u_j)$, the model produces scores $g_{j,a}$ for the next token. With a positive decoding temperature τ_{dec} , define

$$\pi_j(a) = \frac{\exp(g_{j,a}/\tau_{\text{dec}})}{\sum_{c \in \mathcal{V}} \exp(g_{j,c}/\tau_{\text{dec}})}, \quad \tau_{\text{dec}} > 0. \quad (\text{D7})$$

Lower temperature magnifies score differences and concentrates the distribution; higher temperature makes it flatter. Temperature changes the sampling distribution, not the learned scores themselves. The next identifier u_{j+1} is selected from π_j by the declared decoding rule and appended to the prefix:

$$(u_1, \dots, u_j) \mapsto (u_1, \dots, u_j, u_{j+1}). \quad (\text{D8})$$

The enlarged prefix is processed again for the following token. During ordinary inference Θ^* stays fixed; the prefix, hidden states, and attention coefficients change. Detokenization assembles the generated identifiers into the final character string.

Online adaptation would be a separate learning procedure and lies outside the present scope.

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